

CHAPTER 10

BASIC NUTRITIONAL FACTORS AFFECTING HEALTH

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The author would like to acknowledge the significant contribution of Marie Spano to this chapter.

After completing this chapter, you will be able to

- know when to refer an athlete to the appropriate resource, such as a medical doctor or a sports dietitian;
- identify the protein, carbohydrate, and fat recommendations for athletes;
- list the dietary recommendations for disease prevention and overall health; and
- list hydration and electrolyte guidelines for different age groups and scenarios and help athletes develop an individualized hydration plan.

Good nutrition provides athletes with the nutrients necessary for general health, growth, development, and repairing and building muscle tissue, as well as the energy needed to train, compete, and maintain mental focus and concentration. A nutrition plan that is tailored to an athlete's specific needs can help decrease risk of injuries and illness and maximize training adaptations (improvements

made from training) while helping the athlete reach his or her performance goals. This chapter focuses on the scientific rationale for performance-enhancing nutrition practices while providing suggestions that will help readers apply sports nutrition science to real-life scenarios.

Despite the plethora of credible nutrition information easily available online, knowledge among NCAA collegiate athletes is poor and does not meet current recommendations (135). In addition, nutrition guidelines created for the general public are not necessarily applicable to athletes. Because each athlete's nutrition needs depend on many factors (age; body size and composition; sex; genetics; environmental training conditions; injuries; medical nutrition needs; and duration, frequency, and intensity of training), nutrition requirements can vary tremendously between athletes, even if they play the same position. Finally, because nutrition is a complex, constantly evolving science, it is important for strength and conditioning professionals to have some basic nutrition knowledge in addition to a list of nutrition professionals they can refer athletes to for individualized nutrition advice based on the latest science.

Role of Sports Nutrition Professionals

Sports nutrition is a complex multidisciplinary field, and athletic trainers, strength and conditioning professionals, physicians, exercise scientists, and food service providers have varying degrees of nutrition knowledge. Job responsibilities of player development staff should be outlined based on the nutrition education and knowledge of the staff member, the type of nutrition information provided, and state licensure laws regarding nutrition practice.

All sports nutrition professionals should be able to answer basic nutrition questions (for example, “What are some healthy snack ideas?”). However, athletes with complex nutrition issues should be referred to the appropriate resource, either a team physician or sports dietitian. The team physician is responsible for overseeing the athlete’s medical care while the sports dietitian is responsible for providing individualized dietary advice. A **sports dietitian** is a registered dietitian (also referred to as a *registered dietitian nutritionist*) with specific education and experience in sports nutrition. The Academy of Nutrition and Dietetics (AND) Board Certified Specialist in Sports Dietetics (CSSD) certification distinguishes registered dietitians with expertise in sports nutrition from other registered dietitians who specialize in other areas of nutrition (see the [sidebar](#) on this topic). Though some sports dietitians have complementary skills or training and may therefore be social workers, athletic trainers, or chefs, a comprehensive sports nutrition program requires full-time attention; therefore, these secondary skills should be considered a complement to the sports dietitian’s knowledge rather than be used to try to fill two distinct job positions with one individual. Finally, higher-level positions generally include a sports dietitian with a master’s degree, PhD, or both. A sports dietitian can help athletes make the connection between plate and performance.

What Is a Board Certified Specialist in Sports Dietetics (CSSD)?

According to the Commission on Dietetic Registration, the credentialing agency for the Academy of Nutrition and Dietetics, a CSSD “provides individual and group/team counseling and education to enhance the performance of competitive and recreational athletes.” Their responsibilities include:

- Counseling individuals and groups on daily nutrition for performance and health
- Translating the latest scientific evidence into practical sports nutrition recommendations
- Tracking and documenting outcomes of nutrition services, serving as a food and nutrition resource for coaches, support staff, and families

Some of the tasks that an individual with a CSSD certification can conduct according to the CSSD exam content outline include:

- Interpreting body composition assessments on active individuals
- Developing appropriate body composition goals
- Estimating total energy expenditure
- Assessing fluid and electrolyte balance in training, performance, and recovery
- Evaluating and educating on dietary supplements, beverages, ergogenic aids using evidence-based analyses for training, performance, hydration, and recovery
- Contributing to collaborative interventions for and monitoring of disordered eating to guide sport participation

Adapted from Commission on Dietetic Registration, “Specialty Practice Experience,” accessed August 22, 2025, <https://www.cdrnet.org/specialty-practice-experience>.; Commission on Dietetic Registration, “CSSD Examination Content Outline,” last modified September 21, 2023, <https://www.cdrnet.org/vault/2459/web/CDR%20CSSD%20ECO%202023%20-%20effective%2011-2023.pdf>.

At times, physicians and sports dietitians may work together to help athletes with eating disorders, nutrient deficiencies, or specific disease states such as diabetes. Because an athlete’s nutrition and medical information may be shared with other members of the player development or coaching staff or family members in order to provide comprehensive care, in the United States all staff should follow Health Insurance Portability and Accountability Act (HIPAA) guidelines for handling an athlete’s protected health information.

A sports nutrition coach is a professional who is not a registered dietitian but has basic training in nutrition and exercise science. For example, the strength and conditioning professional can act as a sports nutrition coach, providing basic nutrition education and suggestions. More complex situations in which food or nutrition is being used to treat or manage a medical condition (that includes a nutrient deficiency) require medical nutrition therapy and fall under the role of the sports dietitian. Sports nutrition coaches may obtain additional education by getting a sports nutrition certification. For example, the National Academy of Sports Medicine (NASM) has a Certified Sports Nutrition Coach certification designed for personal trainers, health coaches, group fitness instructors, and health care professionals. Also, the International Society of Sports Nutrition (ISSN) Sports Nutrition Specialist (SNS) Certification requires a high school diploma for the exam and is geared toward personal trainers or other fitness professionals who do not hold a four-year college degree. The ISSN also has a Certified Sports Nutritionist credential (CISSN), which requires a four-year undergraduate degree in a related field; or be practicing in the field of sports nutrition for at least five years; or be certified as a personal trainer by the NSCA, ACE, or ACSM; and is intended for health, fitness, and medical professionals who work with athletes and active individuals.

A sports nutritionist with an advanced degree is a professional who may work in the sports nutrition industry or conduct research in the area of sports nutrition and would therefore be able to discuss the literature on a particular topic. The sports nutritionist with an advanced degree may also choose to obtain a sports nutrition certification. One option is the IOC (International Olympic Committee) Diploma in Sports Nutrition. This two-year online program includes coursework, seminars, tutorials, and laboratory practical work. The IOC states that students interested in this course typically have a degree in nutrition or dietetics, biological sciences (including biochemistry, physiology, or sport science), or medicine.

All sports nutrition professionals must follow state nutrition licensure laws, which vary from state to state and specify who is allowed to provide individualized nutrition counseling and medical nutrition therapy. For instance, in Louisiana, general nutrition education can be provided by various disciplines if the information is general, accurate, and not individualized (based on a specific person's dietary needs). However, only a licensed dietitian or nutritionist can provide nutrition assessment and counseling. Nutrition counseling is defined as "the provision of the individual guidance on appropriate food and nutrient intake for those with special needs, taking into consideration health, cultural, socioeconomic, functional, and psychological facts from the nutrition assessment. Nutrition counseling may include advice to increase or decrease nutrients in the diet; to change the timing and size of and composition of meals; to modify food textures, and in extreme instances, to change the route of administration" (www.lbedn.org).

Many individuals without any (or with minimal) education in nutrition and exercise science and little to no formal training refer to themselves as sports nutritionists. Regardless of the specific title of the person providing sports nutrition information or individualized dietary advice, strength and conditioning professionals should take a close look at the person's education (including the curriculum), previous work history (especially the person's day-to-day job duties), knowledge of sports nutrition, and years of experience.

Experienced sports dietitians help athletes make the connection between plate and performance. They have advanced knowledge, skills, and expertise in sports nutrition.

The first step in nutrition coaching involves defining the athlete's goals and identifying the coach's goals (since the two may be different). After this, similar to the strength and conditioning professional undertaking a needs analysis, the sports dietitian will

take a detailed look at the athlete's diet, individual food preferences (including cultural and religious considerations), cooking skills, access to food, financial constraints, barriers to making wise dietary choices, supplement use, weight and body composition history, medical history, training program, and injuries. The sports dietitian will then work with the athlete to develop a plan that fits his or her lifestyle and taste preferences while including the appropriate calorie level; macronutrients and micronutrients in recommended amounts; adequate fluids and electrolytes; and supplements as necessary to help correct a nutrient deficiency, make up for potential nutrient shortfalls, or meet training goals.

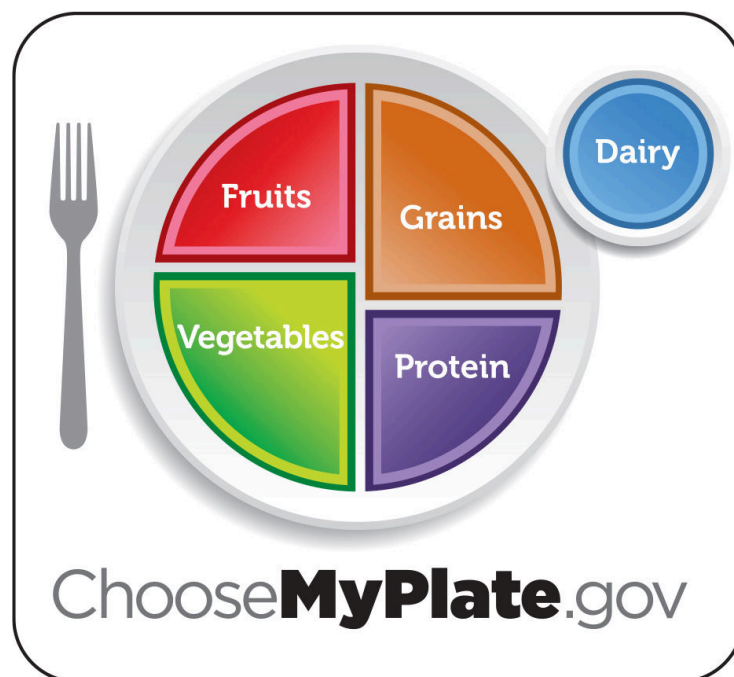


FIGURE 10.1 The MyPlate icon.

U.S. Department of Agriculture

Standard Nutrition Guidelines

For general nutrition information, strength and conditioning professionals may want to refer their athletes to **MyPlate**, a food guidance system created by the U.S. Department of Agriculture in 2011 and based on the Dietary Guidelines for Americans to help consumers make better food choices (100). MyPlate icon illustrates the five food groups based on a mealtime visual, the place setting ([figure 10.1](#)). Another helpful resource geared more toward athletes is the Performance Enhancing Plate from the Collegiate and Professional Sports Dietitians Association (CPSDA, www.sportsrd.org). This plate encourages athletes on hard training days to make half their plate whole grains to fuel muscles, followed by 1/4 of the plate from fruits and vegetables and the other 1/4 from proteins.

MyPlate Guidelines and Recommendations

Information related to MyPlate can be found at www.myplate.gov. Although the basic guidelines represented by the MyPlate icon and corresponding educational materials are universal, they include calorie guidelines and portion recommendations for fruits, vegetables, grains, protein foods, and dairy and fortified soy alternatives, based on age and sex for individuals who get less than 30 minutes of moderate physical activity most days, as noted in [tables 10.1](#) to [10.6](#) on [pages 221](#) through [226](#). Those who are more physically active should adjust the guidelines to meet their specific dietary needs (143). Additional key nutritional topics for MyPlate include guidelines for oils, added sugars, saturated fats, sodium, and alcohol. Oils are not a food group but provide important nutrients such as mono- and polyunsaturated fats and vitamin E. Added sugars are sugars and syrups that are added to foods or beverages when they are made. To build healthy eating habits and stay within calorie needs, people over age 2 should eat foods and drink beverages with little or no added sugars; athletes who need hydration beverages with added sugars are the exception to this

recommendation (3). Saturated fat is fat that is often found solid at room temperature. Examples of these include milk fat, butter, or the fat inside or around meat. Ideally, saturated fat should be limited in the diet and the focus should be on unsaturated fats such as liquid oils used in cooking. In a general, nonactive population, sodium should also be minimized in the diet to no more than 2,300 milligrams per day, but health recommendations for lower salt intake may not be appropriate for endurance athletes and those who sweat a lot. Alcohol has calories but few nutrients; adults of legal drinking age who choose to drink should do so in moderation.

MyPlate should be considered a starting point that athletes can use to evaluate their diet. In general, if a diet provides a variety of foods from each group, it is more likely to contain an adequate amount of each macronutrient, vitamin, and mineral. However, if the diet excludes an entire food group, specific nutrients may be lacking. For example, an athlete who excludes dairy from the diet may have a tough time meeting his or her nutrition requirements for calcium and vitamin D (in fortified milk and fortified yogurts). Fortified soy alternatives are viewed by the USDA as the only plant-based milk alternative equivalent to milk, whereas other nondairy milk alternatives do not have the same nutritional value as milk and must be chosen carefully to avoid deficiencies in calcium and vitamin D (45). Added sugars are also a concern in plant-based milk alternatives. Individuals who exclude all animal foods and fish might not meet their vitamin B₁₂ needs. B₁₂ is found in meat, poultry, fish, eggs, and dairy foods, though some breakfast cereals, nondairy milk alternatives, meat substitutes, and nutritional yeast are fortified with B₁₂. Thus, it is highly recommended that individuals who exclude food groups work with their sports dietitian to find suitable replacements or replacement combinations to ensure that they meet their nutrient requirements for health and performance.

Foods within each food group share similar nutrient compositions and are considered interchangeable; however, a variety of foods should be consumed within each group. For example, eating an

orange, an apple, and a pear provides a broader array of essential nutrients than is provided by three apples. A diet providing a variety of foods from each group is more likely to meet a person's **macronutrient** (carbohydrate, protein, and fat) as well as **micronutrient** (vitamins and minerals) needs.

The resources section of the MyPlate website offers several tools to help plan a healthy diet. MyPlate Plan gives food group targets of what and how much to eat within a person's calorie allowance based on age, sex, height, weight, and physical activity level. It also comes as an app, MyPlate App, to help build healthy habits, pick daily food goals, and set real-time progress. Further, there is the Shop Simple with MyPlate Tool that can be used to find cost-saving food-purchasing methods and new ways to prepare budget-friendly foods.

Dietary Reference Intakes

Because athletes eat food, not individual nutrients, dietary recommendations should be presented in terms of specific food choices. However, it is also important to have an understanding of the athlete's nutrient requirements in order to make food recommendations. The **Dietary Reference Intakes (DRIs)**, developed by the U.S. Department of Agriculture (USDA) and of Health and Human Services (HHS), are a complete set of nutrient intakes for use when evaluating and planning diets for healthy individuals. The DRIs are listed for the macronutrients as well as micronutrients, electrolytes, and water (71-75, 109, 118, 164). The DRIs are based on the body of literature regarding nutrient intake and the reduction of chronic disease, as opposed to simply prevention of dietary deficiencies (115). Because nutrient intake can vary considerably from day to day, the DRIs apply to a person's usual intake. Therefore, when a sports dietitian assesses a person's nutrient intake, they look at several days in order to get an average daily intake for each nutrient. This is especially true with regard to evaluating intake of nutrients found in few foods or in very small amounts in foods (115). Fewer days' worth of food records are

needed to get a good estimation of average protein intake, since this does not vary much from day to day (115). The DRIs include the following:

Recommended Dietary Allowance (RDA)—The average daily nutrient requirement adequate for meeting the needs of nearly all (97%-98%) healthy people within each life stage and sex; often used to plan nutritionally adequate diets for people.

Adequate Intake (AI)—The average daily nutrient intake level recommended when a RDA cannot be established.

Tolerable Upper Intake Level (UL)—The maximum average daily nutrient level not associated with any adverse health effects. Intakes above the UL increase potential risk of adverse effects. (The UL represents intake from all sources including food, water, and supplements.)

Estimated Average Requirement (EAR)—The average daily nutrient intake level considered sufficient to meet the needs of half of the healthy population within each life stage and sex.

The Dietary Guidelines for Americans 2020-2025 has identified several dietary nutrients that are under-consumed by the public (males and females in all age groups); these nutrients are identified as the Nutrients of Public Health Concern and include dietary fiber, vitamin D, calcium, and potassium (56). Dietary patterns that do not meet the recommended intakes of fruits, vegetables, and whole grains are often lacking in adequate dietary fiber and potassium. More than 90 percent of women and 97 percent of men do not meet recommended intakes for dietary fiber. Potassium-rich foods include vegetables, fruits, dairy, and protein foods, particularly white baked potatoes, acorn squash, cooked spinach, cooked plantains, kiwifruit, 100% orange juice, bananas, and plain nonfat or low-fat yogurt. Dairy foods, fortified beverages (e.g., soy beverages and orange juice), and canned sardines and salmon are excellent sources of calcium, while fatty fish, fortified beverages (e.g., milk, orange juice, soy beverages), and fortified yogurt are excellent sources of vitamin

D. Also, iron is a concern for specific populations, particularly female athletes (149). Iron-deficient anemia and iron deficiency are five to seven times more common in female athletes than male athletes. Shellfish (e.g., oysters and mussels), iron-fortified cereals, dark leafy greens (e.g., spinach), and soybeans are excellent sources of iron, providing 20% or more of the daily value for iron per serving; red meat and poultry are considered good sources (41a).

For More Information on Nutrition

Strength and conditioning professionals can rely on the following websites for more information about nutrition:

DRI Calculator for Healthcare Professionals (based on the DRIs, this dietary planning tool calculates daily nutrient recommendations):

www.nal.usda.gov/human-nutrition-and-food-safety/dri-calculator

National Institutes of Health—Office of Dietary Supplements:

<https://ods.od.nih.gov>

The Dietary Supplement Label Database (DSLDD) from the Office of Dietary Supplements (ODS) at the National Institutes of Health (NIH) captures all information printed on labels of dietary supplement products sold in the United States and can be used by consumers to learn more about ingredients listed within the Supplement Facts label by clicking on ingredient names:

<https://dsldd.od.nih.gov>

The Healthy Living and Weight section of the USDA Nutrition.gov website gives information and support for establishing and maintaining a positive body image through healthy eating habits, strategies for success to help someone to lose or gain weight safely and effectively, weight management for children and teens, and information on how to evaluate claims made by weight-loss products and diets:

www.nutrition.gov/topics/healthy-living-and-weight

Peer-reviewed journal articles:

www.pubmed.gov

Collegiate and Professional Sports Dietitians Association:

www.sportsrd.org

International Society of Sports Nutrition:

www.sportsnutritionistsociety.org

Sports and Human Performance Nutrition (SHPN):

<https://www.shpndpg.org/home>

Macronutrients

A macronutrient is a nutrient that is required in significant amounts in the diet. Three important classes of macronutrients are protein, carbohydrate, and fat.

Protein

Protein is the primary structural and functional component of every cell in the human body (164). Dietary proteins are used for growth and development and to build and repair cells; they also serve as enzymes, transport carriers, and hormones. Therefore, dietary protein intake is essential for maintaining health, reproduction, and cellular structure and function (164).

Proteins are composed of carbon, hydrogen, oxygen, and nitrogen. *Amino* means “nitrogen containing,” and **amino acids** are the molecules that, when joined in groups of a few dozen to hundreds, form the thousands of proteins occurring in nature. Proteins in the human body are composed of various combinations of the 20 individual amino acids. Nine amino acids are “essential” because the body cannot manufacture them—they must be obtained through the diet. Four amino acids can be synthesized by the human body and are therefore considered “nonessential” because they do not need to be consumed in the diet. Finally, seven amino acids are considered conditionally essential. These amino acids are typically not essential but become essential, and therefore must be obtained through the diet, during times of illness and stress. The amino acids are listed in [table 10.7](#).

TABLE 10.7 Essential, Nonessential, and Conditionally Essential Amino Acids

Essential	Nonessential	Conditionally essential
Histidine	Alanine	Arginine

Based on Institute of Medicine Standing Committee on the Scientific Evaluation of Dietary Reference Intakes, Food and Nutrition Board (2005), 593.

Essential	Nonessential	Conditionally essential
Isoleucine	Asparagine	Cysteine (cystine)
Leucine	Aspartic acid (aspartate)	Glutamine
Lysine	Glutamic acid (glutamate)	Glycine
Methionine		Proline
Phenylalanine		Serine
Threonine		Tyrosine
Tryptophan		
Valine		

Based on Institute of Medicine Standing Committee on the Scientific Evaluation of Dietary Reference Intakes, Food and Nutrition Board (2005), 593.

Amino acids are joined together by peptide bonds. Two amino acids together are referred to as a *dipeptide*, and several amino acids together are referred to as a **polypeptide**. Polypeptide chains bond together to form a multitude of proteins with various structures and functions. Almost half of the body's protein reserve exists as skeletal muscle, while approximately 15% makes up structural tissues including skin and blood. The rest of the body's protein reserves are in visceral tissues such as the liver and kidney and in the bones (47).

Protein Quality and Dietary Recommendations

Protein quality is determined by amino acid content and **protein digestibility**, as calculated by how much of the protein's nitrogen is absorbed during digestion and its ability to provide the amino acids necessary for growth, maintenance, and repair. Higher-quality proteins are highly digestible and contain all of the essential amino acids. Animal-based proteins—including eggs, dairy foods, meat, fish, and poultry—contain all of the essential amino acids, while soy, quinoa, and buckwheat are the only plant-based proteins that contain all nine essential amino acids. In general, plant proteins are less digestible than animal proteins, though digestibility can sometimes be improved through processing and preparation (110, 164). Although measures of protein quality, such as the **protein digestibility correct amino acid score (PDCAAS)**, take protein

digestibility (or **bioavailability**) into account as well as a protein's ability to provide the essential amino acids necessary for the synthesis of body proteins and other metabolites, they do not take into account how other compounds within the food alter the bioavailability of the protein's amino acids. Foods contain antinutritional factors—compounds that reduce the digestion and absorption of a nutrient, making it less available for use by the body (92). Several antinutritional factors lead to digestive losses and structural changes of amino acids that limit amino acid bioavailability (110). For instance, during cooking, some foods will brown; this browning, called a *Maillard reaction*, leads to compounds that can decrease the bioavailability of certain amino acids. Although the majority of plant-based foods fall short in one or more essential amino acids, all people, not just vegetarians and **vegans** (those who consume only plants and plant products—no meat, fish, poultry, eggs, milk, or other foods that come from animals), can meet their protein needs by consuming a variety of plant foods including legumes, vegetables, seeds, nuts, rice, and whole grains that provide different amino acids so that all essential amino acids are consumed over the course of the day (152). Novel plant-based meat alternatives are also an emerging option to meet dietary protein needs, particularly given that our world population is growing and living longer, and animal proteins alone cannot sustain all of our protein needs.

While dietary recommendations are stated as protein requirements, the actual requirement is for amino acids. The need for amino acids in sedentary, healthy adults results from the constant turnover of cells and cellular proteins. During cell turnover—the constant breakdown and regeneration of cells—the body's free amino acid pool is the immediate and largest supplier of amino acids (105). The pool is replenished from dietary protein digestion, as well as the amino acids released from tissue turnover. Substantially more protein is turned over daily than is ordinarily consumed, indicating that amino acids are recycled (119). This process is not completely

efficient, however, so dietary amino acid intake is required to replace losses.

The RDA for protein for men and women 19 years of age and older, based on nitrogen balance studies, is 0.8 grams of high-quality protein per kilogram body weight per day (164). Children, teens, and pregnant and lactating women have higher dietary protein needs as reflected in the RDA for these groups. However, protein needs are inversely proportional to calorie intake because small amounts of protein can be metabolized as a source of energy when a person is in a state of negative calorie balance, when fewer calories are consumed than are expended (e.g., typically only 1%-6% of total caloric expenditure in most circumstances but up to 10% during prolonged exercise in a glycogen-depleted state) (98, 164). In this case, the protein cannot be used for the intended purpose of replacing the amino acid pool. When caloric intake goes down, protein requirement goes up (101). Therefore, the Institute of Medicine (IOM) established an **Acceptable Macronutrient Distribution Range (AMDR)** for protein, which covers a wide range of protein intake. The AMDR is 5% to 20% of total calories for children ages 1 to 3 years, 10% to 30% of total calories for children ages 4 to 18 years, and 10% to 35% of total calories for adults older than age 18 years (164). Adult men and women typically consume an average of 16% of their calories from protein (68). The AMDR for a nutrient includes a range of intakes associated with reduced risk of chronic disease while still providing adequate intakes of other essential nutrients (118). Though the DRI for protein fits within the AMDR, the DRI is based solely on body weight and therefore does not take low or high calorie intakes into account. Based on the AMDR, when calorie intake is lower, protein needs to go up as a percentage of total calorie intake by approximately 1% for every 100-calorie decrease below 2,000 calories. When total calorie intake is higher, protein needs, expressed as a percentage of total caloric intake, go down to a certain point. In practice, sports dietitians

should first establish an athlete's protein intake and then add carbohydrate and fats as determined by total calorie needs (95).

Concerns About the RDA for Protein

Controversy surrounds the RDA for protein since some scientists suggest that adults should consume more than the RDA for bone health (59), weight management, and building and repairing muscle (122, 123). In addition, research suggests that higher-protein, lower-carbohydrate diets can favorably affect blood lipids, particularly in obese individuals, and therefore also decrease some of the risk factors for cardiovascular disease and metabolic syndrome (94).

Protein is a building block for strong bones, contributing to 50% of bone volume and 33% of bone mass (96). Protein's effect on bone may be due, in part, to its influence on insulin-like growth factor-1 (IGF-1), which is produced in the liver and promotes bone and muscle formation (144). Though research suggests that supplemental, but not dietary, protein increases calcium losses through urine, in healthy individuals consuming 0.7 to 2.1 grams of protein per kilogram body weight, both urinary calcium excretion and intestinal calcium absorption increase (88). In fact, low dietary protein intake (0.7 g protein per kg body weight per day) suppresses intestinal calcium absorption (87). The effects of protein on bone health have been the center of some controversy, but experimental studies generally indicate that dietary protein supports bone health when consumed as part of a high-quality whole-food diet containing adequate calcium. However, future high-quality experimental studies are necessary to fully understand the interactive effects of dietary protein and diet quality on bone health (37).

Protein also plays a multifaceted role in weight management. First, it promotes satiety in a dose-dependent manner: Greater amounts of protein lead to a greater increase in satiety (48). However, the satiating effect of protein also depends on the timing of protein intake, form (solid vs. liquid), concurrent intake of other

macronutrients, and time until the next meal (4). The type of protein may also have an effect on satiety, though research has not fully elucidated which proteins may have the greatest impact (16, 101). Protein also has the greatest thermic effect of feeding—more calories are burned during the digestion of protein as compared to carbohydrate or fat. Finally, higher protein diets help spare muscle loss while a person is on a reduced-calorie diet (93, 132).

The amino acids in protein are used for growth (including muscle growth) and for repairing tissue, synthesizing enzymes and hormones, and repairing and making new cells. Though adults in a general fitness program can likely meet their protein requirements by consuming 0.8 to 1.0 gram of protein per kilogram body weight per day, athletes and those who exercise intensely require more protein (76). Traditionally, aerobic endurance athletes have placed less of a priority on protein in comparison to carbohydrate; however, adequate protein intake and timing are crucial for optimal performance. Protein recommendations for aerobic endurance athletes are 1.2 to 2.0 grams of protein per kilogram body weight per day (167). For building muscle mass and for maintaining muscle mass through a positive muscle protein balance, an overall daily protein intake in the range of 1.4 to 2.0 grams of protein per kilogram body weight per day is ideal (76). Athletes who are on a reduced-calorie diet may need more protein per day to preserve muscle tissue during weight loss (76).

In addition to eating the right amount of protein per day, research supports the idea of athletes consuming sufficient protein as soon as possible after exercise, when muscle tissue is most receptive to amino acids in order to maximize muscle growth (76). In fact, after exercise, both **muscle protein synthesis** and breakdown are increased, though net protein balance is negative when the exercise is done in a fasted state (127). Protein consumed after exercise increases muscle protein synthesis, and muscles remain sensitive to protein ingestion for at least 24 hours following a resistance training bout (114). Reviews surrounding postexercise protein ingestion have

suggested that when recommended levels of protein are consumed, the effect of protein timing appears to be, at best, minimal; total daily protein intake is the most important factor in promoting exercise-induced muscle growth (5, 140). While nutrient timing of protein potentially can be a beneficial strategy for enhancing muscle protein synthesis, the effects on muscle hypertrophy ultimately depend on the timing of the preworkout protein meal; the closer the meal is to the training session, the less need for immediate postworkout protein intake (140). To maximize muscle growth, consume high-quality protein (at a dose of approximately 0.4 to 0.5 g of protein per kg of lean body mass) both before and after exercise within about 4 to 6 hours of each other, depending on meal size (140). Athletes with high body masses may struggle to consume enough protein to meet required daily needs. Therefore, it is recommended that these athletes consume high-quality protein immediately to 2 hours after exercise (86). Ingesting 20 to 40 grams of high-quality protein every 3 to 4 hours throughout the day appears to most favorably affect muscle protein synthesis and is associated with improved body composition and performance outcomes, particularly in strength athletes (86). Aerobic endurance athletes may benefit from the addition of high-quality protein during endurance exercise in addition to their regular carbohydrate intake to suppress muscle damage and improve feelings of soreness (76).

Concerns surrounding protein intakes above the RDA are unfounded for the most part in healthy individuals (105). Proteins consumed in excess of the amount needed for the synthesis of tissue are broken down (103), the nitrogen is excreted as urea in urine, and the remaining ketoacids are either used directly as sources of energy or converted to carbohydrate (**gluconeogenesis**) or body fat (64). In fact, a study examining high protein intake in athletes found that protein intakes up to 2.8 grams per kilogram body weight (as assessed by a seven-day diet record) did not impair any measures of renal function (130). The strength and conditioning professional should be aware that consistently high protein intakes,

above recommended levels for building and repairing muscle, are not recommended because carbohydrate and fat intake (and the nutrients commonly found in carbohydrate- and fat-rich foods) may be compromised. The protein content of commonly consumed foods is listed in [table 10.8](#).

Athletes require more than the RDA for protein to build and repair muscle. Depending on the sport and the training program, 1.2 to 2.0 grams per kilogram body weight of protein is recommended.

Carbohydrate

Carbohydrate primarily serves as a source of energy. However, carbohydrate is not an essential nutrient, because the body can break down the carbon skeletons of certain amino acids and convert them into glucose (i.e., gluconeogenesis). Carbohydrates are composed of carbon, hydrogen, and oxygen. Carbohydrates can be classified into three groups according to the number of sugar (saccharide) units they contain: monosaccharides, disaccharides, and polysaccharides.

Monosaccharides (glucose, fructose, and galactose) are single-sugar molecules. In the body, glucose is present as circulating sugar in the blood, where it is used as the primary energy substrate for cells. In addition, glucose molecules make up glycogen, a polysaccharide stored in muscle and liver cells. In food, glucose is typically combined with other monosaccharides to form various sugars, such as sucrose. Isolated glucose found in candy or sport drinks is found in the form of dextrose, a chemical isomer of glucose. Fructose has the same chemical formula as glucose, but because the atoms are arranged differently, it tastes much sweeter and has different properties. Fructose accounts for the sweet taste of honey and occurs naturally in fruits and vegetables. In the body, fructose causes less insulin secretion than other sugars, which has made it a focus of much research in the area of aerobic endurance

performance. Galactose, the third monosaccharide, combines with glucose to form lactose, milk sugar.

TABLE 10.8 Protein Content of Common Foods

Food	Serving size	Serving size in grams	Average amount of protein (in grams) per serving
Almonds	1/4 cup	39	8
Bacon, pork	3 slices	27.3	11
Beef, ground, 80% lean, pan browned	3 oz	85	23
Black beans	1/2 cup	92.5	7.5
Cashews	1/4 cup	32	5.5
Cheese, feta	1 oz	28	4
Cheese slice, cheddar	1 slice	21	5.5
Cheese slice, Swiss	1 slice	28	7.5
Cheese, string, part skim	1 oz (1 stick)	28	7
Chicken breast	3 oz	85	25
Chicken, thigh with skin	1	111	28
Clams, in shell	3 oz	85	22
Cottage cheese, 1%	4 oz	113	14
Deli turkey	5 oz	142	19
Edamame	1/2 cup	73.5	8.5
Egg, whole, large	1	56	7
Garbanzo beans	1/3 cup	83.5	6
Goat cheese, soft	1 oz	28	5
Greek yogurt, nonfat	6 oz	170	17

Data from U.S. Department of Agriculture, Agricultural Research Service, Beltsville Human Nutrition Research Center. FoodData Central. Available from <https://fdc.nal.usda.gov/>.

Food	Serving size	Serving size in grams	Average amount of protein (in grams) per serving
Hamburger patty	3 oz	85	22
Herring, pickled	¼ cup	35	5
Hot dog, beef	1	50	6
Kidney beans	1/2 cup	88.5	7.5
Kielbasa, beef polska	2 oz	56	7
Lamb chop	3 oz	85	24
Lima beans	1/2 cup	94	7
Lobster, steamed	3 oz	85	16
Mahimahi fish, baked	3 oz	85	20
Milk, 1%, low-fat	8 fl oz	245	8
Mussels, steamed	3 oz	85	20
Oysters, raw	4 oz	113	6.5
Peanut butter	2 tbsp	32	8
Peanuts	1/4 cup	35.5	9
Pistachios	1/4 cup	30.5	7
Pork, ham	3 oz	85	15
Pork tenderloin	3 oz	85	23
Salmon, canned	2 oz	56	13
Salmon, filet, baked	3 oz	85	17
Sausage links	3 links	63	10
Shrimp, large	4	22	5
Sirloin steak	3 oz	85	26
Soy milk	8 fl oz	245	8
Soy nuts	1/4 cup	23	9

Data from U.S. Department of Agriculture, Agricultural Research Service, Beltsville Human Nutrition Research Center. FoodData Central. Available from <https://fdc.nal.usda.gov/>.

Food	Serving size	Serving size in grams	Average amount of protein (in grams) per serving
Tilapia, filet, baked	3 oz	85	22
Tortilla, flour, 8 in.	1	51	4.5
Trout, filet, baked	3 oz	85	20
Tuna, canned	3 oz (1/2 can)	85	22
Turkey breast, roasted	3 oz	85	25
Turkey, ground cooked	3 oz	85	23
Turkey, sausage smoked	1 oz	28	4
Wheat bagel, 5 in.	1 bagel	98	10
Wheat bread	1 slice	29	3
Veal steak	3 oz	85	21
Venison roast	3 oz	85	25
Whey protein powder isolate	1 oz	30	20
Yogurt, Greek, plain, low-fat	1 cup	245	24
Yogurt, plain, low-fat	1 cup	245	13

Data from U.S. Department of Agriculture, Agricultural Research Service, Beltsville Human Nutrition Research Center. FoodData Central. Available from <https://fdc.nal.usda.gov/>.

Disaccharides (sucrose, lactose, and maltose) are composed of two simple sugar units joined together. Sucrose (or table sugar), the most common disaccharide, is a combination of glucose and fructose. Sucrose occurs naturally in most fruits and is crystallized from the syrup of sugar cane and sugar beets to make brown, white, or powdered sugar. Lactose (glucose + galactose) is found only in mammalian milk. Maltose (glucose + glucose) occurs primarily when polysaccharides are broken down during digestion. It also occurs in

the fermentation process of alcohol and is the primary carbohydrate in beer.

Polysaccharides, also known as complex carbohydrates, contain up to thousands of glucose units. Some of the most common polysaccharides of nutritional importance are starch, fiber, and glycogen. *Starch* is the storage form of glucose in plants. Grains, legumes, and vegetables are good sources of starch. Before starch can be used as a source of energy, it must be broken down into glucose components. Dietary fiber, a constituent of the plant cell wall, is also a form of carbohydrate. Cellulose, hemicellulose, beta-glucans, and pectins are fibers; these and noncarbohydrate fibrous materials (lignins) are partially resistant to human digestive enzymes. Fibers have different physiological effects in the body. Some delay gastric emptying, which may temporarily influence feelings of fullness, while other types of fiber increase bulk and water content, reducing constipation and decreasing transit time of feces. In addition, some soluble fibers decrease the absorption of cholesterol and may therefore help reduce blood cholesterol levels after ingestion (104), while prebiotic dietary fibers selectively stimulate the growth of bacteria in the gut (33, 133). Fiber-rich foods include beans, peas, bran, many fruits and vegetables, and some whole-grain foods.

Glycogen is found in small (80, 161) amounts in human liver and muscle, totaling approximately 15 grams of glycogen per kilogram body weight (1) and in animal tissue as a temporary source of stored energy (22). While present in animal meats we eat such as steak, chicken breast, and fish fillets, it is not present to any large extent. When glucose enters the muscles and liver, if it is not metabolized for energy it can be synthesized to form glycogen. Three-quarters of the glycogen in the body is stored in skeletal muscle; the remaining quarter is stored in the liver (22). The process of converting glucose to glycogen is called **glycogenesis**.

Athletes typically consume a variety of carbohydrates in their normal diets. However, when they want to improve the quality of

their diet, by consuming fewer foods that have untoward effects on overall health or performance, they may consider macronutrient ranking systems such as the glycemic index or glycemic load.

Glycemic Index and Glycemic Load

The **glycemic index (GI)** ranks carbohydrates according to how quickly they are digested and absorbed, and therefore raise blood glucose levels, in the 2-hour time period after a meal, compared to the same amount (by weight in grams) of a reference food, typically white bread or glucose, which is given a GI of 100 (78).

Glycemic index = Incremental area under the curve for the blood glucose response after consumption of a 25 or 50g carbohydrate portion of a test food ÷ Incremental area under the curve for the blood glucose response after consumption of the same portion (in grams) of a standard food × 100 (10.1)

Low-GI foods are digested and absorbed slowly, resulting in a smaller rise in blood glucose and subsequent insulin release from the pancreas compared to the reference food (7, 78). Insulin helps lower blood glucose levels by facilitating glucose transport into cells. The fate of glucose within cells depends on where it is shuttled. For instance, muscle cells use glucose for energy while fat cells convert glucose into triglycerides (fat).

Though the GI was developed to help people better control their blood sugar levels—something that is particularly helpful for those with diabetes—research in large cohorts of participants has suggested that diets with high versus low GI foods are associated with increased incidence of type 2 diabetes, cardiovascular disease, diabetes-related cancers, and all-cause mortality (49, 79). However, the GI system has a few issues that may limit its accuracy. First, published GI values for a given type of food may vary considerably due to differences in testing and variations in ingredients used; ripeness of food; and method of food processing, cooking, and storage (28, 62). Secondly, consuming carbohydrate as part of a meal or in different quantities affects the GI (7, 8). Low-GI foods

generally include vegetables, legumes, beans, and whole grains ([table 10.9](#)).

TABLE 10.9 Glycemic Index (GI) of Various Foods

Low-GI foods (<55)	Medium-GI foods (55-69)	High-GI foods (>70)
Apple juice, unsweetened	Brown rice, boiled	Cornflakes cereal
Banana, ripe	Carrots, boiled	English muffin
Carrots, raw	Couscous	Glucose
Chocolate, milk and dark	Honey	Jellybeans
Corn tortilla	Muesli	Potato, boiled
Grapefruit	Pineapple, raw	Potato, instant mashed
Ice cream	Popcorn	Rice crackers, crisps
Kidney beans	Potato, French fries	Rice milk
Lentils	Rice vermicelli	Shredded wheat
Milk, soy or dairy	Rolled oats, quick	Tofu-based frozen dessert
Orange, raw	Soft drink, soda	Watermelon, raw
Quinoa, cooked	Spaghetti, pasta, boiled	White bread
Tomato juice	Sweet potato, boiled	White rice, boiled
Yogurt, fruit	Tropical blend fruit drink	Whole wheat bread

Based on Atkinson et al. (2021).

Since the GI indicates how quickly a carbohydrate-rich food affects blood glucose levels, it is hypothesized that eating foods with different GI values before, during, and after exercise can affect exercise performance (111). Consumption of a low-GI meal before exercise can lead to a reduced blood glucose and insulin response, which has been proposed to help promote fat oxidation and maintain normal blood glucose levels during exercise (156). Although some scientists have speculated that consumption of low-GI foods before exercise might spare carbohydrate by minimizing insulin secretion and therefore improve performance, evidence to support this hypothesis is inefficient (111). The research is mixed, with some studies showing that consumption of a low-GI food as compared to a high-GI food before exercise improves exercise time to exhaustion (41, 162) and others finding that consuming a preexercise low-GI food versus a high-GI food had no effect on running and aerobic

endurance exercise performance (29, 155). Therefore, athletes who use the GI to guide their food choices while balancing health and performance can maintain an adequate amount of dietary carbohydrate with low-GI foods as the base of their choices while sticking with high-GI foods in and around exercise training to provide immediate sources of energy (sugar) for activity (25) and to more rapidly replenish glycogen stores (86).

The **glycemic load** (GL) takes the amount of carbohydrate, in grams, in a portion of food into account—a factor that also influences glycemic response. Because GL takes portion size into account, it is a more realistic gauge of glycemic response than the GI, which is based on a standard serving size. [Table 10.10](#) shows the difference in GI and GL for specific foods.

The GL is equal to the GI multiplied by the amount of carbohydrate in a portion of the food and dividing this total by 100.

$$\text{Glycemic load} = \text{GI of an individual food} \times \text{grams of carbohydrate per serving of food} \div 100 \quad (10.2)$$

Foods with a higher GL are expected to lead to greater increases in blood sugar and subsequent insulin release (7, 8). A low-GL diet, combined with exercise, has been shown to improve insulin sensitivity in older, obese adults and therefore may be a potential treatment for this population (90), while observational and intervention studies suggest that low-GI and low-GL diets are associated with lower levels of inflammatory markers and fasting insulin (79). Chronic low-grade inflammation is considered a potential risk factor for chronic diseases (21). In addition, some (though not all) studies have found that diets with a lower GI or GL are associated with lower levels of fasting insulin as well as fewer cardiovascular risk factors (67, 145).

Fiber

Diets low in **fiber** have been associated with constipation, heart disease, colon cancer, and type 2 diabetes. The DRI for fiber ranges from 21 to 25 grams per day for women (depending on age,

pregnancy, and lactation) and 30 to 38 grams per day for men based on age group. Fiber is commonly found in fruits, vegetables, nuts, seeds, legumes, and whole-grain products such as whole-grain bread, oatmeal, and popcorn.

Health Factors Associated With Food Sensitivities and Allergies

A food allergy is defined as “an adverse health effect arising from a specific immune response that occurs reproducibly on exposure to a given food,” and a food sensitivity, also known as *food intolerance*, is defined as “nonimmune reactions that include metabolic, toxic, pharmacologic, and undefined mechanisms” (17). Reactions to food allergies range in severity from rash to anaphylaxis, which is life-threatening (151). The eight most common food allergens in the United States are cow’s milk, wheat, eggs, tree nuts, peanut, fish, shellfish, and soy (53); these foods must be identified on ingredient labels to prevent adverse reaction in affected individuals (10). Celiac disease (CD) is a type of food allergy triggering an immune response to dietary gluten, a protein found in cereals such as wheat, rye, and barley (97). Food intolerances, especially self-reported, are more prevalent than food allergies (53); a United States–based survey among adults showed that the prevalence of self-reported food intolerances was 25% (77). The most common food intolerance is to lactose, affecting nearly 65% of the population (53). Some food intolerances cannot be readily explained and are classified as undefined; a well-known example of this is irritable bowel syndrome (IBS), characterized by abdominal pain, bloating, abdominal distension, flatulence, constipation, or diarrhea (151). Dietary management of food sensitivities often includes elimination diets, where common trigger foods are removed from the diet, and then slowly reintroduced (151). Individuals with food allergies and intolerances should work with a registered dietitian to ensure nutritional needs are met.

Fat

Although the terms **fat** and *lipid* are often used interchangeably, *lipid* is a broader term. Lipids include **triglycerides** (fats and oils) as well as related fatty compounds, such as sterols and phospholipids. The lipids of greatest significance in nutrition are triglycerides, **fatty acids**, phospholipids, and cholesterol. Triglycerides are formed by the union of glycerol with three fatty acids. The majority of lipids

found in foods and in the body are in the triglyceride form, and within this chapter the term *fat* refers to triglycerides.

TABLE 10.10 Glycemic Index and Glycemic Load

Food	Serving size (g)	Glycemic index (GI; glucose = 100)	Available carbohydrates (g/serving)	Glycemic load (GL; per serving)
Angel food cake	50	67 ± 5	29	19
Apple (different varieties)	120	38 ± 2	15	6
Apple juice, unsweetened	(250 mL) 8.5 fl oz	40 ± 1	29	12
Bagel, white, frozen	70	72	35	25
Banana, ripe	120	51	25	13
Blueberry muffin	60	50 ± 3	31	15
Cheese pizza (Pillsbury Canada Ltd)	100	60	27	16
Chocolate milk (powder dissolved in milk)	250 mL (8.5 fl oz)	43	11	5
Chocolate pudding, instant, made from powder and whole milk	100	47 ± 4	16	7
Corn, sweet	150	53 ± 4	32	17
English muffin	30	77 ± 7	14	11
Grapes	120	46 ± 3	18	8
Ice cream	50	61 ± 7	13	8
Jellybeans	30	80 ± 8	28	22
Kidney beans, canned	150	52	17	9

Based on Atkinson et al. (2008).

Food	Serving size (g)	Glycemic index (GI; glucose = 100)	Available carbohydrates (g/serving)	Glycemic load (GL; per serving)
Mashed potatoes, instant	150	85 ± 3	20	17
Muesli	30	55 ± 10	19	11
Oatmeal, prepared	50	69	35	24
Plantain	120	68 ± 6	31	21
Potato chips	50	51 ± 7	24	12
Raisins	60	64 ± 11	44	28
Rice, brown (steamed, USA)	150	50	33	16
Rice, white, long grain, boiled 20-30 min	150	50	36	18
Soft drink	250 mL (8.5 fl oz)	63	26	16
Strawberries	120	40 ± 7	3	1
Sweet potato, boiled	150	61 ± 7	28	17
Tomato juice	250 mL (8.5 fl oz)	23 ± 6	7	2
Vanilla wafers	25	77	18	14
Whole wheat bread	30	72 ± 6	12	8
Yogurt, low-fat (0.9%), wild strawberry	200	31 ± 14	30	9

Based on Atkinson et al. (2008).

Like carbohydrate, fat contains carbon, oxygen, and hydrogen atoms, but because the fatty acid chains have more carbon and hydrogen relative to oxygen, they provide more energy per gram. For example, fats provide approximately 9 kcal/g, while

carbohydrates and protein provide approximately 4 kcal/g. Dietary fats and oils are composed of different types of fatty acids.

Saturated fatty acids have no double bonds, and their carbon molecules are saturated with hydrogen. Saturated fatty acids are used for certain physiological and structural functions, but the body can make these fatty acids; therefore, there is no dietary requirement for saturated fatty acids. Unsaturated fatty acids contain some carbon molecules that are joined together by double bonds, making them more chemically reactive. Fatty acids containing one double bond are **monounsaturated**. With two or more double bonds, a fatty acid is **polyunsaturated**. Two polyunsaturated fatty acids are considered essential, meaning that the body cannot make them: omega-6 and omega-3 fatty acids. These two fatty acids are necessary for the formation of healthy cell membranes, proper development and functioning of the brain and nervous system, and hormone production. Omega-6 fatty acids are abundant in the food supply and found in foods such as soybean, corn, and safflower oil and products made with these oils. Fewer foods contain omega-3 fatty acids; these include fish, particularly fatty fish such as salmon, herring, halibut, trout, and mackerel, which contain the omega-3 fatty acids **eicosapentaenoic acid (EPA)** and **docosahexaenoic acid (DHA)**. Eicosapentaenoic acid and DHA are tied to a dose-dependent decrease in triglycerides; a small, but statistically significant, decrease in blood pressure, especially in the elderly; and potential antiarrhythmic effects (106, 113).

One could also consume flaxseeds, walnuts, soybean oil, or canola oil to meet omega-3 requirements since these contain the **omega-3 fatty acid alpha-linolenic acid (ALA)**, which is converted to EPA and DHA. However, this conversion process is inefficient. According to vivo studies, approximately 5% of ALA is converted to EPA, and <0.5% of ALA is converted to DHA in adults (129). Therefore, while foods that contain ALA count toward one's omega-3 intake, they do not have a substantial effect on EPA and DHA levels in the body. Foods rich in ALA may improve some risk

factors for cardiovascular disease; however, it is not clear which is responsible—ALA, the other compounds in these nutrient-rich foods, or a combination of the two (43). In addition to its physiological functions, fat is important because it is responsible for the characteristic flavor, aroma, and texture of many foods. Generally, most dietary fats and oils are a mix of all three types of fatty acids, with one type predominating. Soy, corn, sunflower, and safflower oils are relatively high in polyunsaturated fatty acids; olive, peanut, and canola oils are high in monounsaturated fatty acids; and most animal fats and tropical oils (e.g., coconut, palm kernel) are relatively high in saturated fatty acids. See [table 10.11](#) for fat content of common foods.

When stored in the human body, fat serves many functions. Energy is stored primarily as adipose tissue in humans, but small amounts are also found in skeletal muscle, especially in aerobically trained athletes (142). Body fat insulates and protects organs, regulates hormones, and carries and stores the fat-soluble vitamins A, D, E, and K.

Relationship With Cholesterol

Cholesterol is a waxy, fat-like substance that is an important structural and functional component of all cell membranes. Additionally, cholesterol is used for the production of bile salts, vitamin D, and several hormones, including the sex hormones (estrogen, androgen, and progesterone) as well as cortisol. Though cholesterol has many essential functions in the body, high levels of cholesterol may lead to atherosclerosis, hardening of the arteries due to plaque buildup on artery walls, which narrows the area within arteries through which blood can pass. Therefore, having high levels of blood cholesterol is a risk factor for heart disease and stroke.

High levels of total cholesterol, **low-density lipoproteins (LDL)**, and triglycerides are all associated with increased risk of heart disease. Low-density lipoprotein is further divided into subfractions based on particle size. Smaller, more dense particles,

called **very low-density lipoproteins (VLDL)**, are more atherogenic (artery clogging) than larger LDL particles (50). High levels of saturated or trans fats, weight gain, and anorexia can all increase LDL cholesterol (159). However, VLDL levels increase with increasing intake of carbohydrate (116). High levels of **high-density lipoproteins (HDL)** are protective against heart disease but are not a target of therapy (practitioners are told not to focus on HDL). [Table 10.12](#) shows how LDL, total, and HDL cholesterol are classified.

High intake of refined carbohydrates, weight gain, excessive alcohol intake, and very-low-fat diets can increase triglycerides (blood fats). However, as with cholesterol, several factors affect triglycerides, including a sedentary lifestyle, overweight or obesity, smoking, genetics, and certain diseases and medications (126, 159).

For optimal cardiovascular health, and risk reduction of chronic diseases, the Scientific Report of the 2020 Dietary Guidelines Advisory Committee recommends limiting saturated fat to less than 10% of total calories and replacing saturated fat with unsaturated fat, particularly polyunsaturated fat. As of June 2018, partially hydrogenated oils (PHOs), the major source of artificial trans fat in the food supply, and a major cause of dyslipidemia, are no longer Generally Recognized as Safe (GRAS). Therefore, PHOs are no longer added to foods. In addition, it is advised that added sugars, particularly from sugar-sweetened beverages, desserts, sweet snacks, candies, and sweetened breakfast cereals, are consumed at a maximum of 10% of total calories (154).

Fat and Performance

Both intramuscular and circulating fatty acids are potential energy sources during exercise. Compared with the limited capacity of the body to store carbohydrate, fat stores are large and represent a vast source of fuel for exercise (20). For example, a lean runner who is 160 pounds (72 kg) with 4% body fat has approximately 22,400 calories stored within fat tissue (57). At rest and during low-intensity

exercise, a high percentage of the energy produced is derived from fatty acid oxidation (137). When the intensity of exercise increases, there is a gradual shift from fat to carbohydrate as the preferred source of fuel. Consistent aerobic training increases the muscle's capacity to use fatty acids (82). In addition to training, the body adapts to using greater amounts of fat for energy when a higher-fat, lower-carbohydrate diet is consumed over a period of time (60, 82). Because the type of diet to which the body is adapted may influence performance (69), the effects of high-fat, low-carbohydrate diets vary, depending on the individual (128, 136, 138).

TABLE 10.11 Fat Content of Common Foods

Food	Serving size	Serving size (g)	Total fat (g)	Saturated fat (g)	Mono-unsaturated fat (g)	Poly-unsaturated fat (g)
Almond butter	1 tbsp	32	17.7	1.3	10.4	4.3
Almonds, whole	22 each	26	13.1	1.0	8.3	3.3
Avocado	1/4 each	50	7.3	1.1	4.9	0.9
Beef, ground, 80% lean, pan browned	3 oz	85	14.8	5.6	6.5	0.4
Beef, sirloin steak	3 oz	85	13.4	5.3	5.6	0.5
Butter, dairy	1 tbsp	14	11.5	7.3	3.0	0.4
Cashews, whole, dry roasted	1/4 cup	34	15.9	3.1	9.4	2.7
Cheese, cheddar, low-fat	1 slice	28	2.0	1.2	0.6	0.1

Data from U.S. Department of Agriculture, Agricultural Research Service, Beltsville Human Nutrition Research Center. FoodData Central. Available from <https://fdc.nal.usda.gov/>.

Food	Serving size	Serving size (g)	Total fat (g)	Saturated fat (g)	Mono-unsaturated fat (g)	Poly-unsaturated fat (g)
Cheese, mozzarella, part-skim, shredded	1/4 cup	28	5.6	3.2	1.6	0.3
Chia seeds	1/4 cup	42	12.9	1.4	1.0	9.9
Chicken, thigh with skin	3 oz	85	12.5	3.5	5.3	2.5
Coconut oil	1 tbsp	14	13.4	11.2	0.9	0.2
Dark chocolate	1 oz	28	9.1	5.4	2.8	0.3
Egg, large	1	50	5.3	1.6	2.0	0.7
Flaxseed, ground	1 tbsp	7	3.0	0.3	0.5	2.0
Goat cheese, soft	1 oz	28	6.0	4.1	1.4	0.2
Margarine, vegetable oil spread 40%	1 tbsp	14	5.4	1.3	1.7	2.2
Mayonnaise, regular	1 tbsp	14	10.3	1.6	2.3	6.2
Milk, skim	1 cup	244	0.2	0.1	0.1	0.0
Milk, whole, 3.25% fat	1 cup	244	7.9	4.6	2.0	0.5
Oil, canola	1 tbsp	14	14.0	1.0	8.9	3.9
Oil, olive	1 tbsp	14	14.0	1.9	9.9	1.4
Oil, safflower, 70% linoleic	1 tbsp	14	14.0	0.8	1.9	10.1
Oil, soybean	1 tbsp	14	14.0	2.1	3.1	7.9

Data from U.S. Department of Agriculture, Agricultural Research Service, Beltsville Human Nutrition Research Center. FoodData Central. Available from <https://fdc.nal.usda.gov/>.

Food	Serving size	Serving size (g)	Total fat (g)	Saturated fat (g)	Mono-unsaturated fat (g)	Poly-unsaturated fat (g)
Peanut butter, natural	2 tbsp	32	16.5	3.3	8.3	4.1
Salmon, filet, wild, baked	3 oz	85	6.9	1.1	2.3	2.8
Tuna, light, canned in water	2 oz	56	0.5	0.1	0.1	0.1
Trout, filet, wild, baked	3 oz	85	5.0	1.4	1.5	1.6
Walnuts, dried, halves	1/4 cup	25	16.3	1.5	2.2	11.8
Yogurt, Greek, plain, low-fat	1 cup	245	4.7	3.0	1.2	0.2
Yogurt, plain, low-fat	1 cup	245	3.8	2.5	1.0	0.1

Data from U.S. Department of Agriculture, Agricultural Research Service, Beltsville Human Nutrition Research Center. FoodData Central. Available from <https://fdc.nal.usda.gov/>.

TABLE 10.12 Classification of LDL, Total, and HDL Cholesterol

LDL cholesterol (mg/dl)	
<100	Optimal
100-129	Near optimal/above optimal

LDL = low-density lipoprotein; HDL = high-density lipoprotein.

Lipoprotein level testing should be completed after a 9- to 12-hour fast.

LDL cholesterol is the primary target of therapy.

Source: National Heart, Lung, and Blood Institute; National Institutes of Health; U.S. Department of Health and Human Services, *Managing Blood Cholesterol in Adults: Systematic Evidence Review from the Cholesterol Expert Panel*, 2013 (U.S. Department of Health and Human Services; National Heart, Lung, and Blood Institute, 2013).

LDL cholesterol (mg/dl)	
130-159	Borderline high
160-189	High
>190	Very high
Total cholesterol (mg/dl)	
<200	Desirable
200-239	Borderline high
>240	High
HDL cholesterol (mg/dl)	
<40	Low
>60	High

LDL = low-density lipoprotein; HDL = high-density lipoprotein.

Lipoprotein level testing should be completed after a 9- to 12-hour fast.

LDL cholesterol is the primary target of therapy.

Source: National Heart, Lung, and Blood Institute; National Institutes of Health; U.S. Department of Health and Human Services, *Managing Blood Cholesterol in Adults: Systematic Evidence Review from the Cholesterol Expert Panel*, 2013 (U.S. Department of Health and Human Services; National Heart, Lung, and Blood Institute, 2013).

The human body has a sufficient amount of fat to fuel long training sessions or competition.

Macronutrient Guidelines

Protein

- Choose a variety of protein foods, from both animal and plant sources, which include seafood, lean meat and poultry, eggs, beans, peas and lentils, soy products, nuts, and seeds.
- Increase the amount and variety of seafood consumed by choosing seafood in place of some meat and poultry.
- Choose nuts, seeds, and soy products in place of some meat and poultry.
- Aerobic endurance athletes should consume 1.2 to 2.0 grams of protein per kilogram body weight per day.

- Strength athletes should consume 1.4 to 2.0 grams of protein per kilogram body weight per day.
- Ingest 20 to 40 grams (0.25-0.40 g of protein per kg of body mass) of high-quality protein every 3 to 4 hours throughout the day.

Carbohydrate

- Reduce intake of calories from added sugars.
- Increase intake of vegetables (prepared without added salt or fat) and fruit (prepared without added sugars).
- Eat a variety of vegetables, including beans; peas; starchy; and dark-green, red, and orange vegetables.
- Consume at least half of all total grains as whole grains. Individuals who eat refined grains should choose enriched grains.

Fats and Alcohol

- Consume less than 10% of calories from saturated fats by replacing them with unsaturated fats, particularly polyunsaturated fats.
- If alcohol is consumed, it should be consumed in moderation—up to one drink per day for women and two drinks per day for men—and only by adults of legal drinking age. Pregnant women should avoid alcohol, and breastfeeding women should be cautious about their intake if they choose to drink and speak to their healthcare provider. Alcohol should be avoided in the time period after exercise because it reduces muscle protein synthesis (124).